# UNIT 7

# FINANCIAL ARITHMETIC

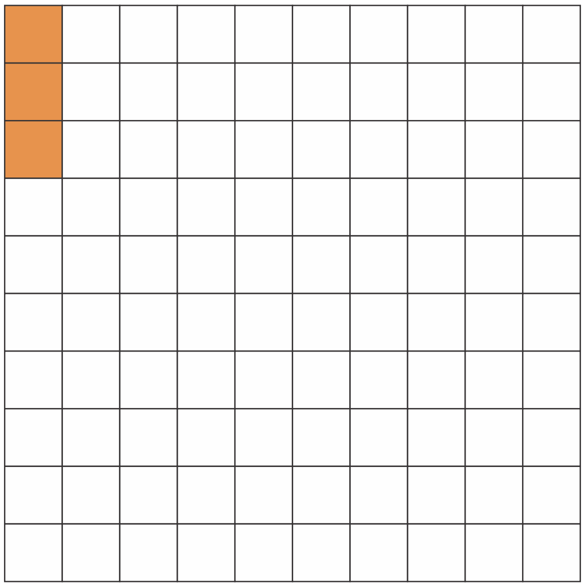
## PERCENTAGE

The word percent is a short form of the Latin word "per centum". Its meaning is "out of hundred".

> Percentage is a special type of fraction which indicates parts out of hundred parts.

**Recognize percentage as a fraction with denominator of 100:**

Look at this figure.



It has been divided into 100 equal parts. 3 out of 100 parts, are colored.

i.e. of the $\frac{3}{100}$ figure is coloured.

We say that "3 percent" of the figure is coloured.

**Convert a percentage to a fraction by expressing it as a fraction with denominator 100 and then simplify:**

**Example 1:** Convert 5% into fraction.

**Solution:**

$$5\% = \frac{5}{100}$$

$$= \frac{\cancel{5}^{1}}{\cancel{100}_{20}} = \frac{1}{20}$$

**Example 2:** Convert 72% in fraction.
**Solution:**

$$72\% = \frac{72}{100}$$
$$= \frac{\cancel{72}^{18}}{\cancel{100}_{25}} = \frac{18}{25}$$

**Example 3:** Convert $7\frac{1}{2}\%$ in fraction.
**Solution:**

$$7\frac{1}{2}\% = \frac{15}{2}\% = \frac{15}{2} \times \frac{1}{100}$$
$$= \frac{15}{200} = \frac{3}{40}$$

**Convert a fraction to a percentage by multiplying it with 100%:**

**1.** The fraction having denominators as 100 can easily be converted into percentage.

**For example:** $\frac{5}{100} = 5\%$ and $\frac{97}{100} = 97\%$

**2.** The fractions having denominators other than 100 are converted into percentage by two ways:

**1.** First convert the fraction into equivalent fraction having denominator as 100 and then express as percentage.

**2.** Convert any fraction into percentage by multiplying and dividing with 100 or multiplying by 100%.

**Example 1:** Convert $\frac{1}{2}$ into percentage.
**Solution:** By converting into equivalent fraction with 100 denominator.

$$\frac{1}{2} = \frac{1 \times 50}{2 \times 50} = \frac{50}{100} = 50 \times \frac{1}{100} = 50\% \; (\because \frac{1}{100} = 1\%)$$

By multiplying with 100%

$$\frac{1}{2} = \frac{1}{2} \times \frac{100}{100}$$

$$= \frac{100}{2} \times \frac{1}{100} = 50\% \ (\because \frac{1}{100} = 1\%)$$

**Example 2:** Convert $\frac{3}{5}$ into percentage.
**Solution:** By converting into equivalent fraction.

$$\frac{3}{5} = \frac{3 \times 20}{5 \times 20} = \frac{60}{100} = 60\%$$

By multiplying with 100%

$$\frac{3}{5} = \frac{3}{5} \times \frac{100}{100}$$

$$= \frac{3 \times 100\%}{5}$$

$$= 60\% \qquad (\because \frac{1}{100} = 1\%)$$

**Example 3:** Convert $\frac{5}{8}$ into percentage.
**Solution:**

$$\frac{5}{8} = \frac{5}{8} \times \frac{100}{100}$$

$$= \left( \frac{5}{8} \times 100 \right) \frac{1}{100} = \frac{125}{2} \% = 62.5$$

**Convert a percentage to a decimal by expressing it as a fraction with denominator 100 and then as a decimal:**

**Example 1:** Convert 3% into decimal.
**Solution:** $3\% = \frac{3}{100} = 0.03$

**Example 2:** Convert 5/2% into decimal.
**Solution:** $\frac{5}{2}\% = 2.5\% = \frac{2.5}{100} = 0.025$

**Example 3:** Convert 135% into decimal.
**Solution:** $135\% = \frac{135}{100} = 1.35$

**Convert a decimal to percentage by expressing it as a fraction with denominator 100 then as a percentage:**

**Example 1:** Convert 0.5 into percentage.

**Solution:**

$$0.5 = \frac{5}{10} = \frac{5 \times 10}{10 \times 10} = \frac{50}{100} = 50\ \%$$

**Example 2:** Convert 0.25 into percentage.

**Solution:**

$$0.25 = \frac{25}{100} = 25\ \%$$

**Example 3:** Convert 0.125 into percentage.

**Solution:**

$$0.125 = \frac{125}{1000}$$

$$= \frac{12.5}{100}$$

$$= 12.5\ \%$$

**Finding percentage of a given quantity:**

**Example 1:** Find 15% of 75.

**Solution:**

$$15\% \text{ of } 75 = \frac{15}{\cancel{100}_{4}} \times \cancel{75}^{3}$$

$$= \frac{45}{4}$$

$$= 11\ \frac{1}{4}$$

**Example 1:** Is 20% of 140 greater than 25% of 150.

**Solution:**

$20\% \text{ of } 140 = 140 \times \frac{\overset{1}{\cancel{20}}}{\underset{5}{\cancel{100}}} = 28$

$25\% \text{ of } 150 = 150 \times \frac{\overset{1}{\cancel{25}}}{\underset{4}{\cancel{100}}} = \frac{75}{2} = 37\frac{1}{2}$

As $28 < 37\frac{1}{2}$

Therefore, 20% of 140 is not greater than 25% of 150.

## EXERCISE 7.1

**Q1. Convert the following into fractions and decimal.**

**(i)** 25% **(ii)** 1% **(iii)** 31% **(iv)** 15% **(v)** $8\frac{1}{2}\%$

**(vi)** 20.5% **(vii)** 175% **(viii)** 115%

**Q2. Convert the following fractions into percentage.**

**(i)** $\frac{1}{4}$ **(ii)** $\frac{7}{20}$ **(iii)** $\frac{8}{25}$ **(iv)** $\frac{5}{4}$

**Q3. Write the following percentage into decimals.**

**(i)** 25% **(ii)** 31% **(iii)** 1% **(iv)** $2\frac{1}{2}\%$ **(v)** 6.5%

**Q4. Convert the following decimal into percentage.**

**(i)** 0.025 **(ii)** 0.4 **(iii)** 0.85

**(iv)** 0.105 **(v)** 12.5 **(vi)** 125.5

**Q4. Cacuate the following decimal into percentage.**

**(i)** 10% of 50 **(ii)** 30% of 70

**(iii)** 25% of 250 **(iv)** 50% of 50

**(v)** 75% of 750

**(vi)** 125% of 50 cm

**(vii)** 60% of 300 litres

**(viii)** 20% of Rs 13.25

**Q6. Find the value of the percentage of each number.**

**(i)** 10% of 70 = ____________

**(ii)** 65% of 20 = ____________

**(iii)** 15% of 400 = ____________

**(iv)** 63% of 100 = ____________

**(v)** 10% of 50 = ____________

**(vi)** 22% of 50 = ____________

**(vii)** 75% of 60 = ____________

**(viii)** 30% of 90 = ____________

**Q7. Solve the following.**

**FRACTION** → Divide numerator by denominator. → **DECIMAL** → Multiply by 100. → **PERCENT**

| FRACTION | DECIMAL | PERCENT |
|---|---|---|
| $\frac{1}{2}$ | | |
| $\frac{3}{4}$ | | |
| $\frac{2}{5}$ | | |
| $\frac{1}{8}$ | | |
| $\frac{7}{8}$ | | |

**Example 1:** Naheed has Rs 120. She spends 25% of the amount. How much money is left with her?

**Solution:**

Total amount = Rs 120

Rate of expenditure = 25% of the total amount

Amount spends = Rs 120 × $\frac{25}{100}$ = 30

Amount left with Naheed = Rs 120 - 30 = Rs 90

**Example 2:** 75% students out of 40 were present in a test. How many students were absent?

**Solution:**

Total students = 40

Percentage of present students = 75%

Number of students present = 75% of 40

= 40 × $\frac{75}{100}$ = 30 students

Number of students absent = 40 - 30 = 10 students

**Example 3:** Find the quantity if 75% of it is 123.

**Solution:** By using the formula.

Since, $\boxed{\text{Percentage} = \frac{\text{Part of quantity}}{\text{Quantity}} \times 100\%}$

Therefore, Quantity = $\frac{\text{Part of quantity}}{\text{Percentage}} \times 100\%$

$= \frac{123}{75\%} \times \frac{100}{100} = \frac{123}{0.75}$

$= \frac{12300}{75} = 164$

**Example 4:** In an examination Akram obtained 510 marks out of 850. Find the percentage of the marks obtained by him.
**Solution:** This question can be solved by the following formula.

$$\text{Percentage} = \frac{\text{Part of quantity}}{\text{Quantity}} \times 100\%$$

Here, Quantity = 850 marks
Part of quantity = 510 marks

So, Percentage $= \frac{510}{850} \times 100\% = 60\%$

## WORD PROBLEMS

**1.** I obtained 60 marks out of 75 marks. What is the percentage of my marks?

**2.** Saima spent Rs 300 out of Rs 500. Find percentage of her expenditure.

**3.** There were 8500 voters in a village. 34% did not cast their votes. Find the number of voters who cast their votes.

**4.** The population of a village is 15000. If the population increases by 5% in a year, find the population after one year.

**5.** Rahila pays 5% of her salary in charity in a month. If she pays Rs 200, find her monthly salary.

**6.** The 18% of the distance between two cities is 36 km.
Find the distance between two cities.

## PROFIT, LOSS AND DISCOUNT

**Selling price**
The amount which is received by selling goods is called "Selling price".

**Cost price**
The amount which is paid to purchase goods is called "Cost price".

**Profit Percentage and Loss Percentage:**
If in a business selling price of an item is more than cost price then there is a profit in such transaction. But if selling price of an item is less than cost price then there is a loss in such transaction.

Thus
**Profit: Selling Price (SP) - Cost Price (CP)**
**Loss: Cost Price (CP) - Selling Price (SP)**

$$\text{Profit Percent} = \frac{\text{Profit}}{\text{CP}} \times 100\%$$ and $$\text{Loss Percent} = \frac{\text{Loss}}{\text{CP}} \times 100\%$$

**Discount:**
The reduction made on the list price or marked price of an article is called **discount** i.e. Net Selling price = Marked Price - Discount.

**Remember:** The discount percent is calculated only on the marked price and not on selling or cost price.

**Example 1:** A school bag is purchased for Rs 180 and is sold out for Rs 225. Find the profit percent.
**Solution:**
Profit = selling price - cost price
= 225 - 180
= 45 rupees

$$\text{Profit Percent} = \frac{\text{Profit}}{\text{CP}} \times 100$$

$= \frac{\cancel{45}^{5}}{\cancel{180}_{\cancel{20}_{1}}} \times \cancel{100}^{5} = \frac{25}{1} = 25\%$

Profit = 25%

**Example 2:** A fruit seller purchased 40 dozens bananas in all and their cost price is Rs 50 per dozen. He sells all the bananas at the rate of Rs 60 per dozen. Find his total profit and profit percent.

**Solution:**

Bananas = 40 dozen
Cost price per dozen = 50 rupees
Total cost price = 50 × 40 = Rs 2000
Selling price per dozen = 60 rupees
Total selling price = 60 × 40 = Rs 2400

Profit = selling price - cost price
= 2400 - 2000 = Rs 400

Profit Percent = Profit × 100

$= \frac{\cancel{400}^{1}}{\cancel{2000}_{5}} \times 100$

$= \frac{1}{\cancel{5}_{1}} \times \cancel{100}^{20} = 20\%$

**Example 3:** Saima purchased a mobile set for Rs 15000. After some time she sold it for Rs 12000. Find her loss percent.

**Solution:**

Here CP = 15000 rupees
SP = 12000 rupees

Since SP < CP, so there is loss

Now Loss = CP - SP
= 15000 - 12000 = Rs 3000

Loss Percent = $\frac{\text{Loss}}{\text{CP}} \times 100$

$$= \frac{\overset{1}{\cancel{3000}}}{\underset{5}{\cancel{15000}}} \times 100$$

$$= \frac{1}{\underset{1}{\cancel{5}}} \times \overset{20}{\cancel{100}}$$

$$= 20\%$$

## EXERCISE 7.2

**Q1. Find the profit or loss percent in the following.**

**i.** Eggs are purchased at Rs 120 per dozen and are sold out at Rs 130 per dozen.
**ii.** A man purchases pencils at Rs 60 per dozen and sells at Rs 70 per dozen.
**iii.** A milkman purchases milk at the rate of Rs 75.50 per litre and sells at the rate of Rs 80 per litre.
**iv.** A shopkeeper purchases 25 breads at Rs 10 per bread. He sells breads at Rs 13 per bread 7 breads are spoiled.
**v.** A man purchases 10 dozen eggs at the rate of Rs 120 per dozen. 3 dozen eggs are broken. Rest of the eggs are sold at the rate of Rs 130 per dozen

**Q2.** A book seller purchases 20 books of Mathematics at Rs. 120 per book and sells at Rs 125 per book. Find his profit and profit percent.

**Q3.** A grocer purchases 10 kilograms of rice at Rs 120 per kilogram and sells at Rs 125 per kilogram. Find profit percent.

**Q4.** A shopkeeper bears 15% loss on the sale of a jacket. If the cost of the jacket is Rs 1000, find his loss.

**Q5.** Umaima bought a dinner set for Rs 52,000 at 20% discount. Find the actual price of the dinner set.